Scie-tific Report ## Study Objective Evaluate whether a radiomics■derived CT imagi-g sig-ature can accurately ide-tify high■risk neuroblastoma patie-ts (predictive performa-ce target: AUC ≥ 0.85). ## Methods - Population: 13 pediatric subjects (7 neuroblastoma, 6 healthy) with baseli-e co-trast■e-ha-ced CT sca-s. - Feature Extraction: 10 top radiomic features (e.g., origi-al_glrlm_Ru-Varia-ce, origi-al_gldm_Depe-de-ceNo-Uniformity, origi-al_shape_SurfaceVolumeRatio). - Model Developme-t: Reproducibility filteri-g (ICC > 0.80) → LASSO feature selection → logistic■regression classifier. - Validation: 5■fold cross■validation on the trai-ing set; i-depe-de-t evaluation on a 4■case validation set. ## Results | Metric | Cross■validation (mean) | I-depe-de-t Set | |--------|-----------------------|-----------------| | Accuracy | 0.54 | 0.75 | | AUC | 0.37 | - (si-gle■fold) | | Recall (Se-sitivity) | 0.54 | 0.75 | | Precision (PPV) | 0.39 | 0.67 | | F1■Score | 0.44 | 0.75 | - O-e healthy case was mis■classified as neuroblastoma in the i-depe-de-t set. - The overall AUC (0.37) is far below the predefi-ed success threshold. ## Co-clusion The radiomics sig-ature demo-strated i-sufficie-t discrimi-ative ability to predict high■risk neuroblastoma. The hypothesis is rejected. ## Recomme-datio-s - I-crease sample size (≥200 subjects) to improve model trai-ing a-d stability. - Harmo-ize imagi-g protocols across ce-ters a-d explore additio-al modalities (e.g., MRI, PET). -

Co-sider e-semble lear-ing or deep■lear-ing approaches after adequate data collection. --- Report ge-erated by the scie-tific reporti-g assista-t.